Cohomological Obstructions to Global Semantic Descent

Event Horizons, Black Holes, and the Failure of Invariant Structure

Adrian Diamond

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Abstract

General Relativity localizes semantic structure by restricting invariance to local domains. In this paper, we show that black holes arise precisely as obstructions to the extension of local semantic regimes into a global invariant structure.

Observational outcomes form local equivalence classes under admissible transformations, but in the presence of horizons, these local regimes fail to admit a global quotient. We formalize this failure as an obstruction to semantic descent and identify the event horizon as the boundary across which invariant structure cannot be globally extended.

Black holes are therefore not singularities of physical substance but boundaries of semantic coherence: regions where global meaning fails to exist.

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1 Introduction

Relativity reorganizes physical interpretation through invariance. A quantity possesses observer-independent meaning if and only if it descends to an invariant under the admissible transformations of the theory.

In Special Relativity, admissible transformations are global (the Lorentz group), and invariant structure is preserved across the entirety of spacetime. This yields a single global semantic regime: all observers agree on the invariant content of physical quantities.

In General Relativity, however, curvature localizes admissible transformations. Coordinate systems and symmetry structures are defined only over local domains, and invariant quantities are correspondingly defined only relative to these domains. As a result, semantic structure becomes inherently local: meaning is no longer globally uniform, but depends on the region of spacetime under consideration.

This localization raises a fundamental structural question:

Under what conditions can locally defined invariant structures be extended to a global semantic regime?

In this paper, we show that black holes arise precisely as obstructions to such global extension. More specifically, we demonstrate that event horizons define boundaries across which local semantic regimes cannot be coherently glued into a single global invariant structure.

Our approach is structural and functorial. Observational data is organized into a presheaf over spacetime, and local semantic regimes are defined as quotients under admissible transformations. The existence of a global semantic regime is then equivalent to the existence of a global section compatible with these local identifications.

Failure of such a global section is interpreted as an obstruction to semantic descent. We formalize this obstruction using sheaf-theoretic and cohomological tools, showing that the presence of an event horizon induces a nontrivial obstruction class in the first cohomology group of the system.

This perspective leads to a reinterpretation of black holes:

- Black holes are not singularities of physical substance, but obstructions to global invariant structure.
- Event horizons are not merely causal boundaries, but boundaries of semantic coherence.
- The failure of global agreement across observers is not paradoxical, but structurally necessary.

In this framework, meaning is identified with invariant structure, and the limits of physical interpretation correspond to the limits of global semantic descent.

The paper is organized as follows. In Section 2, we define observational space and local semantic regimes. In Section 3, we formulate the problem of global semantic descent. In Section 4, we introduce the semantic presheaf and establish the obstruction theorem. Section 5 provides interpretation, and Section 6 connects the framework to gauge theory. The Appendix develops the cohomological formulation of the obstruction in detail.

2 Observational Space and Local Semantic Regimes

Let M denote a spacetime manifold, and let $\mathcal{O}$ denote the observational space of the theory. Elements of $\mathcal{O}$ represent observational outcomes, which may include coordinate-dependent measurements, field values, or other representational data.

In General Relativity, admissible transformations are defined only locally. For each open set $U \subset M$, let $\mathcal{O}(U)$ denote the observational outcomes restricted to U, and let $\mathcal{G}(U)$ denote the admissible transformations acting on $\mathcal{O}(U)$.

These transformations induce an equivalence relation on $\mathcal{O}(U)$:

$$x \sim_U y \quad \Longleftrightarrow \quad \exists g \in \mathcal{G}(U) \text{ such that } g \cdot x = y.$$

Definition 2.1 (Local Semantic Regime)**.** *The* local semantic regime *over U is the quotient space*

$$\mathcal{R}(U) := \mathcal{O}(U)/\sim_U .$$

Each $\mathcal{R}(U)$ represents the invariant content of observational data within the domain U. Elements of $\mathcal{R}(U)$ are equivalence classes of observational outcomes under admissible transformations, and therefore encode observer-independent meaning within that domain.

Definition 2.2 (Local Semantic Meaning)**.** *Let S be a set of values. A quantity $Q : \mathcal{O}(U) \to S$ is said to possess* local semantic meaning *on U if it descends to a well-defined function*

$$\widetilde{Q}_U : \mathcal{R}(U) \to S.$$

Equivalently, Q is invariant under the action of $\mathcal{G}(U)$.

Thus, in curved spacetime, meaning is inherently local: it is defined only relative to domains over which admissible transformations act coherently. The passage from $\mathcal{O}(U)$ to $\mathcal{R}(U)$ constitutes a local instance of semantic descent, in which representational data is reduced to invariant structure.

3 Global Semantic Descent

We now consider the problem of extending local semantic regimes to a global invariant structure.

Let $\{U_i\}$ be an open cover of spacetime M. For each U_i, we have a local semantic regime

$$\mathcal{R}(U_i) = \mathcal{O}(U_i)/\sim_{U_i}.$$

A global semantic regime would consist of a single quotient space

$$\mathcal{R} = \mathcal{O}(M)/\sim$$

together with a projection

$$\pi : \mathcal{O}(M) \to \mathcal{R}$$

such that, for each U_i, the restriction of π to $\mathcal{O}(U_i)$ is compatible with the local quotient $\mathcal{R}(U_i)$.

Definition 3.1 (Global Semantic Regime)**.** *A* global semantic regime *exists if there is a quotient $\mathcal{R}$ and a projection $\pi : \mathcal{O}(M) \to \mathcal{R}$ such that, for every open set $U \subset M$, the restriction of π to $\mathcal{O}(U)$ factors through $\mathcal{R}(U)$.*

Equivalently, the following diagram commutes for each U:

$$\begin{array}{ccc} \mathcal{O}(U) & \longrightarrow & \mathcal{R}(U) \\ \big\downarrow & & \big\downarrow \\ \mathcal{O}(M) & \xrightarrow{\pi} & \mathcal{R} \end{array}$$

where the vertical maps are restrictions and the horizontal maps are quotient projections.

Definition 3.2 (Global Semantic Meaning)**.** *A quantity $Q : \mathcal{O}(M) \to S$ possesses* global semantic meaning *if it descends to a well-defined function*

$$\widetilde{Q} : \mathcal{R} \to S.$$

Thus, the existence of global meaning is equivalent to the compatibility of local semantic regimes across all domains of spacetime.

When such compatibility holds, the local quotients $\mathcal{R}(U_i)$ assemble into a single global invariant structure. When it fails, semantic structure remains fragmented across local regimes, and no global semantic regime exists.

Theorem 3.3 (Global Descent Criterion)**.** *A quantity $Q : \mathcal{O}(M) \to S$ possesses global semantic meaning if and only if it is compatible with all local equivalence relations $\sim_{U_i}$ and descends to a well-defined function on R.*

When such a global quotient exists, meaning is globally coherent. When it does not, semantic structure remains fragmented across local regimes.

4 Obstruction to Global Semantic Regimes

We now recast the problem of global semantic descent in sheaf-theoretic terms.

Let $\mathcal{O}$ denote the assignment

$$U \mapsto \mathcal{O}(U),$$

and let $\mathcal{R}$ denote the assignment

$$U \mapsto \mathcal{R}(U) = \mathcal{O}(U)/\sim_U .$$

Restriction maps are induced by inclusion of open sets, so that both $\mathcal{O}$ and $\mathcal{R}$ define presheaves on M.

Definition 4.1 (Semantic Presheaf)**.** *The assignment*

$$\mathcal{R} : \mathrm{Open}(M)^{\mathrm{op}} \to \mathbf{Set}, \quad U \mapsto \mathcal{R}(U),$$

is called the semantic presheaf*.*

A global semantic regime corresponds to a global section of $\mathcal{R}$ compatible with all local identifications.

Definition 4.2 (Global Section)**.** *A* global semantic regime *exists if there is a section*

$$s \in \mathcal{R}(M)$$

whose restrictions agree with the local quotient structures on all open subsets.

Definition 4.3 (Gluing Condition)**.** *The semantic presheaf $\mathcal{R}$ satisfies the* gluing condition *if, for any open cover $\{U_i\}$ and any family of local sections*

$$s_i \in \mathcal{R}(U_i)$$

that agree on overlaps $U_i \cap U_j$, there exists a unique global section

$$s \in \mathcal{R}(M)$$

such that $s|_{U_i} = s_i$ for all i.

Theorem 4.4 (Semantic Obstruction Theorem)**.** *If the semantic presheaf $\mathcal{R}$ fails to satisfy the gluing condition on a cover $\{U_i\}$ of M, then no global semantic regime exists.*

Proof. Failure of the gluing condition implies that compatible local sections cannot be extended to a global section. Therefore, there is no element of $\mathcal{R}(M)$ representing a globally consistent equivalence class of observational outcomes. Hence, no global semantic regime exists. □

Thus, global semantic meaning exists if and only if the semantic presheaf behaves as a sheaf with respect to the admissible transformations.

We now formalize the notion of an event horizon in structural terms.

Definition 4.5 (Event Horizon)**.** *An* event horizon *is a subset $H \subset M$ such that admissible transformations cannot be extended across H to produce a consistent equivalence relation on any open set intersecting both sides of H.*

Theorem 4.6 (Horizon Obstruction Theorem)**.** *Let $H \subset M$ be an event horizon. Then there exists an open cover $\{U_i\}$ of M such that the semantic presheaf $\mathcal{R}$ fails to satisfy the gluing condition across H. In particular, no global semantic regime exists that extends across H.*

Proof. Consider open sets U_1 and U_2 on opposite sides of the horizon H, together with neighborhoods intersecting H. On each U_i, admissible transformations define equivalence relations $\sim_{U_i}$ and corresponding local regimes $\mathcal{R}(U_i)$.

However, by definition of an event horizon, admissible transformations on U_1 and U_2 cannot be extended to a common transformation on any open set intersecting both regions. Therefore, the induced equivalence relations are not compatible on overlaps.

This incompatibility prevents the existence of a consistent family of transition maps between local semantic regimes, and hence obstructs the gluing condition.

Therefore, $\mathcal{R}$ fails to admit a global section compatible with all local identifications, and no global semantic regime exists across H. □

Thus, the event horizon is not merely a causal boundary but a structural boundary: it marks the failure of global semantic descent.

5 Example: Schwarzschild Spacetime and Horizon Obstruction

We illustrate the preceding framework using the Schwarzschild spacetime, which provides the simplest model of a black hole with a well-defined event horizon.

Let M denote the Schwarzschild spacetime with metric

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2,$$

defined on the domain $r > 0$. The surface

$$r = r_s := 2GM$$

defines the Schwarzschild radius, which is the location of the event horizon.

5.1 Local Semantic Regimes

Consider an open cover $\{U_i\}$ of M such that:

- Some sets U_{out} lie entirely in the exterior region $r > r_s$,
- Some sets U_{in} lie entirely in the interior region $r < r_s$,
- Some sets intersect neighborhoods arbitrarily close to the horizon.

On each open set U, admissible transformations $\mathcal{G}(U)$ consist of local diffeomorphisms preserving the geometric structure of the spacetime. These induce local equivalence relations on observational data $\mathcal{O}(U)$, and hence local semantic regimes $\mathcal{R}(U)$.

Within each region (interior or exterior), these transformations act coherently, and invariant quantities (e.g., curvature scalars) define well-behaved elements of $\mathcal{R}(U)$.

5.2 Failure of Global Compatibility

However, the structure of Schwarzschild spacetime introduces a fundamental asymmetry across the horizon:

- In the exterior region $r > r_s$, the coordinate t is timelike and r is spacelike.
- In the interior region $r < r_s$, the roles reverse: r becomes timelike and t becomes spacelike.

This change is not a coordinate artifact but reflects a genuine structural transition in the causal geometry of spacetime.

As a consequence, admissible transformations defined in U_{out} cannot be extended across the horizon to U_{in} in a way that preserves the equivalence relations defining $\mathcal{R}(U)$.

In particular:

- Observational outcomes in U_{out} and U_{in} are organized by fundamentally different transformation structures.
- There is no consistent identification of equivalence classes across sets intersecting both regions.

Thus, the local semantic regimes $\mathcal{R}(U_{\text{out}})$ and $\mathcal{R}(U_{\text{in}})$ cannot be glued into a single global quotient.

5.3 Cohomological Interpretation

We now interpret the failure of global semantic descent in Schwarzschild spacetime in cohomological terms.

Let $\mathcal{G}$ denote the sheaf of admissible transformations on M, where for each open set $U \subset M$, $\mathcal{G}(U)$ consists of the local symmetry transformations acting on $\mathcal{O}(U)$. These transformations preserve the equivalence relations defining the local semantic regimes $\mathcal{R}(U)$.

Let $\{U_i\}$ be an open cover of M adapted to the horizon, including sets in the exterior region ($r > r_s$), interior region ($r < r_s$), and neighborhoods approaching the horizon.

On overlaps $U_i \cap U_j$, the compatibility of local semantic regimes is encoded by transition functions

$$g_{ij} \in \mathcal{G}(U_i \cap U_j),$$

relating the equivalence classes in $\mathcal{R}(U_i)$ and $\mathcal{R}(U_j)$.

These transition functions satisfy the cocycle condition on triple overlaps:

$$g_{ij} \cdot g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k.$$

Thus, the collection $\{g_{ij}\}$ defines a Čech 1-cocycle with values in the sheaf $\mathcal{G}$.

Definition 5.1 (Horizon Obstruction Class)**.** *The* horizon obstruction class *is the cohomology class*

$$[\{g_{ij}\}] \in H^1(M, \mathcal{G}),$$

associated with the transition data of local semantic regimes across the chosen open cover.

Theorem 5.2 (Schwarzschild Horizon Obstruction)**.** *In Schwarzschild spacetime, the obstruction class*

$$[\{g_{ij}\}] \in H^1(M, \mathcal{G})$$

associated with any open cover intersecting the event horizon is nontrivial.

Sketch. Suppose, for contradiction, that the obstruction class is trivial. Then there exist local sections $h_i \in \mathcal{G}(U_i)$ such that

$$g_{ij} = h_i^{-1} h_j \quad \text{on } U_i \cap U_j.$$

This would imply that the local semantic regimes $\mathcal{R}(U_i)$ can be consistently reparameterized so that all transition functions become trivial, yielding a global section of the semantic presheaf $\mathcal{R}$.

However, in Schwarzschild spacetime, the causal structure differs fundamentally across the horizon: the interchange of timelike and spacelike directions implies that no admissible transformation can identify observational data across regions $r > r_s$ and $r < r_s$ in a way that preserves the local equivalence relations.

Therefore, no such family $\{h_i\}$ exists, and the cocycle $\{g_{ij}\}$ is not cohomologically trivial. Hence,

$$[\{g_{ij}\}] \neq 0.$$

□

Corollary 5.3. *The event horizon in Schwarzschild spacetime defines a nontrivial cohomology class in $H^1(M, \mathcal{G})$, and therefore constitutes an intrinsic obstruction to global semantic descent.*

Interpretation. The obstruction class $[\{g_{ij}\}] \in H^1(M, \mathcal{G})$ measures the precise failure of invariant structure to globalize. The event horizon is therefore characterized by a nontrivial cohomology class, encoding the intrinsic impossibility of extending local semantic regimes into a global one.

This establishes that the event horizon is not merely a geometric or causal boundary, but a cohomological obstruction: a structural feature of spacetime encoded in the failure of local symmetry data to globalize.

6 Interpretation

The preceding results admit a direct interpretation.

In Special Relativity, invariance is global and meaning is uniform across all observers. In General Relativity, curvature localizes invariance, and meaning becomes domain-dependent.

Black holes represent the extreme case of this localization: regions where semantic regimes cannot be coherently unified.

This perspective reframes several standard interpretations:

- Singularities do not represent infinite physical quantities, but limits of invariant description.
- The event horizon does not merely obscure information; it marks the boundary beyond which semantic coherence fails.
- Observational disagreement across horizons is not paradoxical, but structurally necessary.

Meaning is not destroyed at a black hole. It becomes irreducibly local.

No contradiction arises because no global semantic regime is required to exist.

7 Connection to Gauge Theory

The semantic structure developed above admits a natural interpretation in the language of gauge theory.

In General Relativity, admissible transformations form a local symmetry groupoid. This is analogous to a gauge symmetry acting on a fiber bundle over spacetime M, where local trivializations correspond to coordinate patches.

Let $\mathcal{O}$ denote the presheaf of observational outcomes. The equivalence relations $\sim_U$ induced by admissible transformations define local identifications analogous to gauge equivalence within each patch U.

Definition 7.1 (Semantic Gauge Structure)**.** *The assignment of local equivalence relations $\{\sim_U\}$ defines a semantic gauge structure on M, in which observational outcomes are identified up to local symmetry.*

In this framework:

- Observational outcomes correspond to sections of a bundle of representations.
- Admissible transformations act as gauge transformations.
- Semantic regimes correspond to gauge-invariant classes.

A global semantic regime exists precisely when these local gauge identifications can be consistently extended across all overlaps, i.e., when the associated bundle admits a global invariant section.

Proposition 7.2. *Failure of global semantic descent is equivalent to the absence of a global gauge-invariant section compatible with all local trivializations.*

Thus, black holes may be interpreted as regions where the semantic gauge structure fails to admit global invariants.

This perspective aligns semantic descent with gauge invariance: both identify physical meaning with quantities invariant under local symmetry.

Appendix A — Cohomological Obstruction to Global Semantic Regimes

We now refine the obstruction to global semantic regimes using cohomological language.

Let $\mathcal{R}$ denote the semantic presheaf assigning to each open set $U \subset M$ the quotient space

$$\mathcal{R}(U) = \mathcal{O}(U)/\sim_U$$

Assume that $\mathcal{R}$ admits restriction maps compatible with inclusions of open sets.

Let $\{U_i\}$ be an open cover of M. On overlaps $U_i \cap U_j$, the local equivalence relations induce transition data describing how local regimes are identified.

Definition 7.3 (Semantic Transition Cocycle). *A semantic transition cocycle is a collection of compatibility data $\{g_{ij}\}$ defined on overlaps $U_i \cap U_j$ describing the relation between local equivalence classes.*

These satisfy the cocycle condition on triple overlaps:

$$g_{ij} \cdot g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k.$$

Definition 7.4 (Sheaf of Admissible Transformations). *Let $\mathcal{G}$ denote the sheaf assigning to each open set $U \subset M$ the group of admissible transformations acting on $\mathcal{O}(U)$. Restriction maps are given by restriction of transformations to smaller domains.*

Definition 7.5 (Obstruction Class). *The obstruction to global semantic descent is represented by a cohomology class*

$$[\{g_{ij}\}] \in H^1(M, \mathcal{G}),$$

where $\mathcal{G}$ is the sheaf encoding admissible transformations.

Theorem 7.6 (Cohomological Obstruction Theorem). *A global semantic regime exists if and only if the semantic transition cocycle $\{g_{ij}\}$ is cohomologically trivial, i.e., represents the zero class in $H^1(M, \mathcal{G})$.*

Sketch. If the cocycle is trivial, then there exist local redefinitions that render all transition data compatible, allowing the construction of a global section. If the class is nontrivial, no such global identification exists, and the local regimes cannot be glued into a single quotient. □

Corollary 7.7. *Nontrivial cohomology corresponds to irreducible fragmentation of semantic structure.*

In the presence of an event horizon, the induced obstruction class is generically nontrivial, preventing the existence of a global semantic regime.

Black holes therefore correspond to nontrivial cohomology classes of the semantic gauge structure, representing intrinsic obstructions to the existence of globally invariant meaning.

References

[1] A. Einstein, *On the Electrodynamics of Moving Bodies*, Annalen der Physik, 1905.

[2] A. Einstein, *The Field Equations of Gravitation*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, 1915.

[3] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.

[4] S. Mac Lane, *Categories for the Working Mathematician*, Springer, 1998.

[5] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Springer, 1982.

[6] R. Hartshorne, *Algebraic Geometry*, Springer, 1977.

[7] J. Baez and J. P. Muniain, *Gauge Fields, Knots and Gravity*, World Scientific, 1994.

[8] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic*, Springer, 1992.